

BIOMECHANICS (EPHE 341)

Electromyography

The electrical signal a contracting muscle leaves behind, and what it will and will not tell you

By the end of this lecture you should be able to...

- **Describe** the physiological basis of the EMG signal, from motor unit to electrode
- **Choose** where to put electrodes, and say what goes wrong at the innervation zone and the tendon
- **Relate** EMG to force, allowing for electromechanical delay and for the fact that the relationship is not proportional
- **Explain** why EMG is recorded as the *difference* between two electrodes
- **Process** a raw signal: band-pass, rectify, RMS or linear envelope, and normalise
- **Interpret** amplitude and frequency changes during fatigue, and know how reliable each one is

What the word means

Electro

electrical

Myo

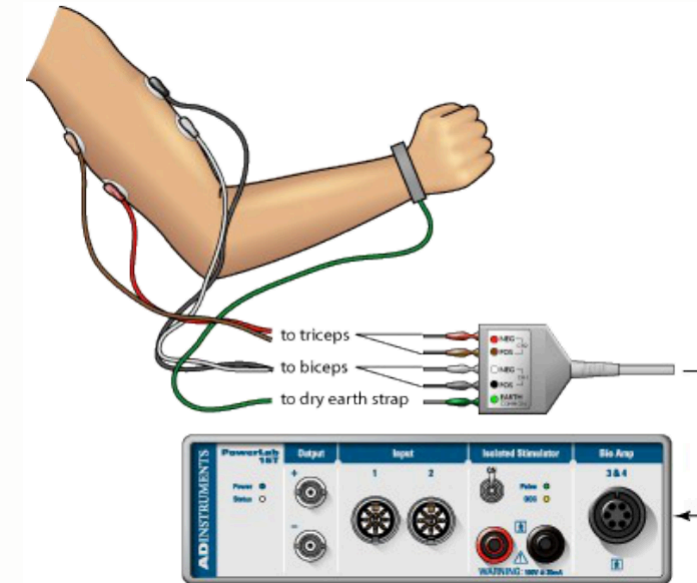
muscle

Graphy

representation,
writing, description

A visual or numeric representation of the electrical activity associated with muscle contraction.

We record, analyse and interpret these signals to understand how the nervous system controls muscle — in health and in disease.



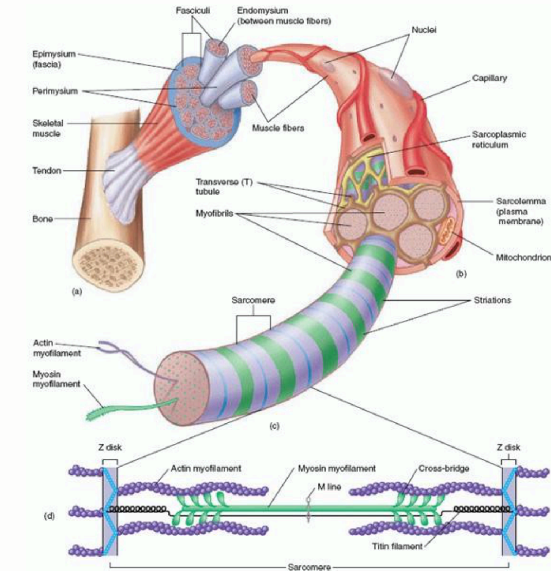
What we need to know

1. Whole muscle **structural**, **physiological** and **neural** organisation
2. The **electrochemical mechanism** of contraction
3. The resulting contraction and the concurrent **electrical signal** collected through EMG electrodes

Three steps, and the signal at the end of them is a long way from the command at the start. Most of the care in using EMG is about remembering what happened in between.

Whole muscle structural organisation

- Whole muscle
- Fascicle
- Muscle fibre
 - myofibril
 - type I, type II
- Sarcomere
 - sliding filament



Muscle physiology — fibre type

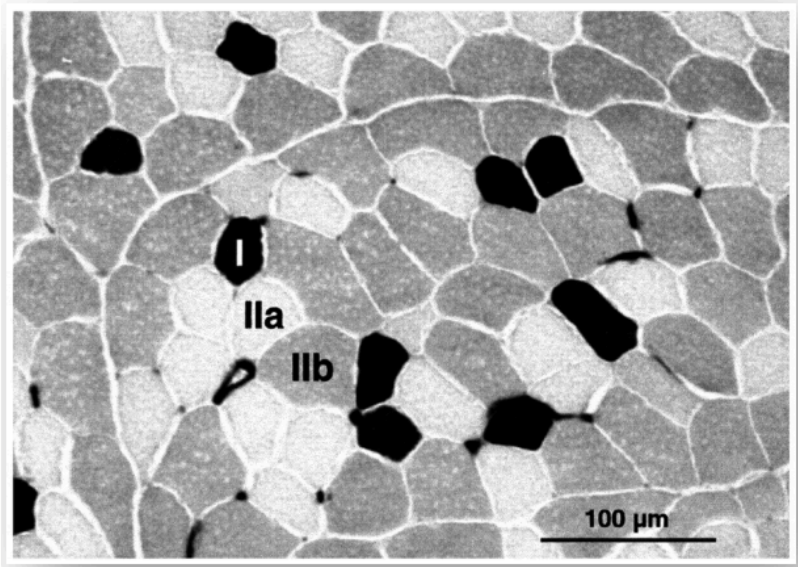


TABLE 9.2 Structural and Functional Characteristics of the Three Types of Skeletal Muscle Fibers			
	SLOW OXIDATIVE FIBERS	FAST OXIDATIVE FIBERS	FAST GLYCOLYTIC FIBERS
METABOLIC CHARACTERISTICS			
Speed of contraction	Slow	Fast	Fast
Myosin ATPase activity	Slow	Fast	Fast
Primary pathway for ATP synthesis	Aerobic	Aerobic (some anaerobic glycolysis)	Anaerobic glycolysis
Myoglobin content	High	High	Low
Glycogen stores	Low	Intermediate	High
Recruitment order	First	Second	Third
Rate of fatigue	Slow (fatigue-resistant)	Intermediate (moderately fatigue-resistant)	Fast (fatigable)
ACTIVITIES BEST SUITED FOR			
	Endurance-type activities—e.g., running a marathon; maintaining posture (antigravity muscles)	Sprinting, walking	Short-term intense or powerful movements, e.g., hitting a baseball
STRUCTURAL CHARACTERISTICS			
Color	Red	Red to pink	White (pale)
Fiber diameter	Small	Intermediate	Large
Mitochondria	Many	Many	Few
Capillaries	Many	Many	Few

Copyright © 2006 Pearson Education, Inc., publishing as Benjamin Cummings.

Muscle physiology — fibre type varies by muscle, and with depth

	type I fibres	type II fibres
Abductor digiti minimi	51.8	48.2
Abductor pollicis brevis	63.0	37.0
Adductor magnus (surface)	53.5	46.5
Adductor magnus (deep)	63.3	36.7
Adductor pollicis	80.4	19.6
Biceps brachii (surface)	42.3	57.7
Biceps brachii (deep)	50.5	49.5
Biceps femoris	66.9	33.1
Brachioradialis	39.8	60.2
Deltoid (superficial)	53.3	46.7
Deltoid (deep)	61.0	39.0
1st Dorsal interosseus	57.4	42.6
Erector spinae (surface)	58.4	41.6
Erector spinae (deep)	54.9	45.1
Extensor digitorum	47.3	52.7
Extensor digitorum brevis	45.3	54.7
Flexor digitorum brevis	44.5	55.5
Flexor digitorum profundis	47.3	52.7
Frontalis	64.1	35.9
Gastrocnemius (lat. head surface)	43.5	56.5
Gastrocnemius (lat. head deep)	50.3	49.7
Gastrocnemius (medial head)	50.8	49.2
Gluteus maximus	52.4	47.6

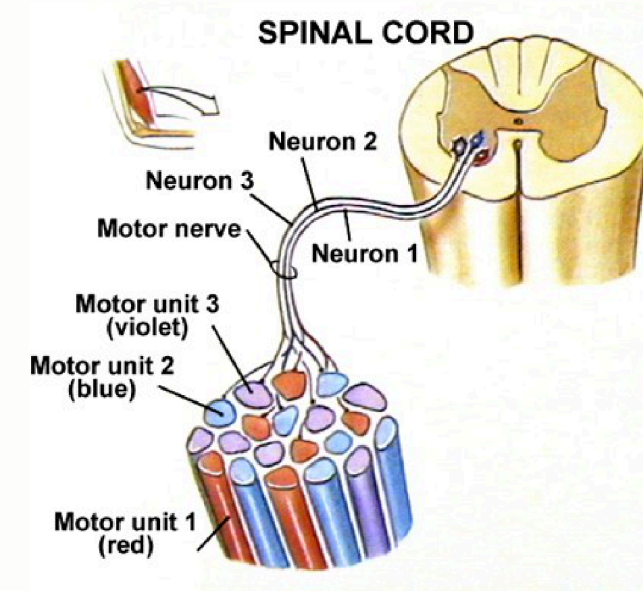
Iliopsoas	49.2	50.8
Infraspinatus	45.3	54.7
Latissimus dorsi	50.5	49.5
Orbicularis oculi	15.4	84.6
Pectoralis major (clavic. head)	42.3	57.7
Pectoralis major (sternal head)	43.1	56.9
Peroneus longus	62.5	37.5
Rectus abdominis	46.1	53.9
Rectus femoris (lat. head surface)	29.5	70.5
Rectus femoris (lat. head deep)	42.0	58.0
Rectus femoris (medial head)	42.8	57.2
Rhomboid	44.6	55.4
Sartorius	49.6	50.4
Soleus (surface)	86.4	13.6
Soleus (deep)	89.0	11.0
Sternomastoid	35.2	64.8
Supraspinatus	59.3	40.7
Temporalis	46.5	53.5
Tibialis anterior (surface)	73.4	26.6
Tibialis anterior (deep)	72.7	27.3
Trapezius	53.7	46.2
Triceps (surface)	32.5	67.5
Triceps (deep)	32.7	19.6
Vastus lateralis (surface)	37.8	62.3
Vastus lateralis (deep)	46.9	53.1
Vastus medialis (surface)	43.7	56.3
Vastus medialis (deep)	61.5	38.5

Johnson et al., *Journal of the Neurological Sciences* **18** (1973) 111–129

Neural organisation of muscle

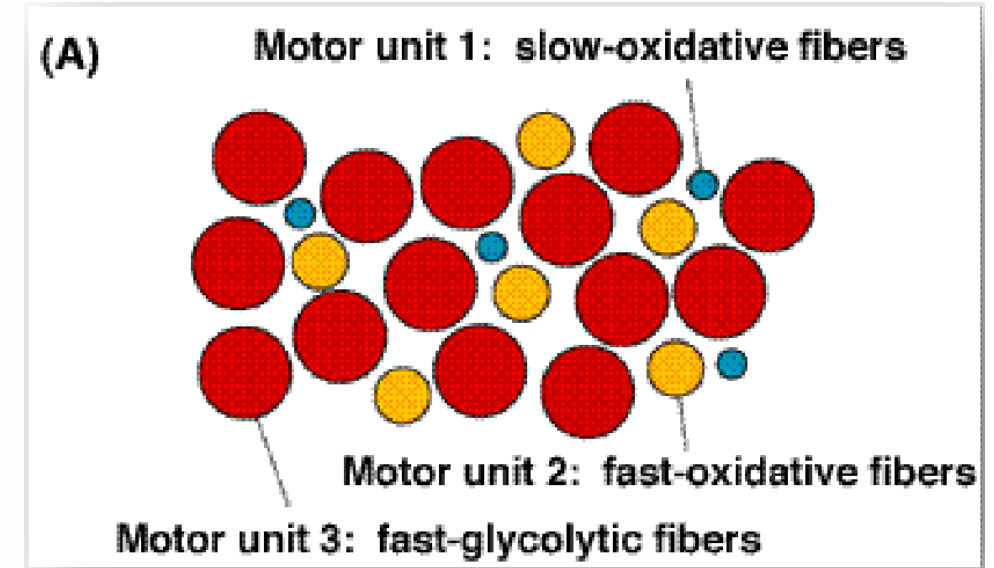
- Each muscle is made up of many units of contraction called **motor units**
- A motor unit is a **motor nerve and all the muscle fibres it innervates**
- A descending command that excites the motor nerve contracts **all** the fibres in that unit, together — an **all-or-none** response

The motor unit, not the fibre, is the smallest thing the nervous system can switch on. It is therefore the smallest thing EMG can see.



Motor units come in sizes

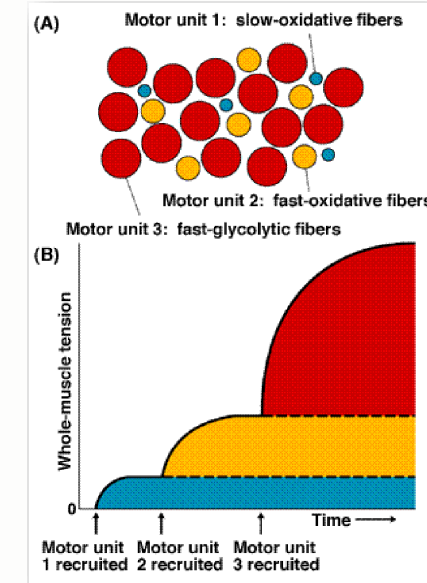
- Motor units can be **small** (few fibres) or **large** (many fibres)
- Within any one motor unit, all fibres are of the **same type** (I, IIa, IIb)
- Most small units are **type I**; most large units are **type II**



Recruitment, and the Henneman size principle

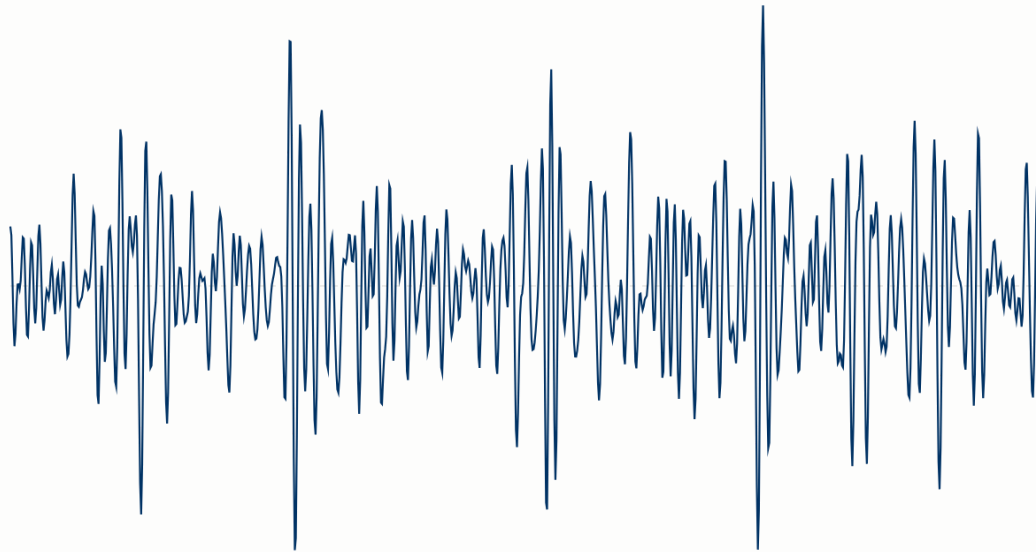
- The nervous system grades contraction by altering **how many** units are recruited and **how often** they fire
- At low firing rates, **small** units are recruited first; at higher rates the **larger** units join in
- Force is increased by adding more active motor units

Orderly recruitment from small to large is the **Henneman size principle**. It gives smooth, non-jerky force production and leans on the fatigue-resistant units first.

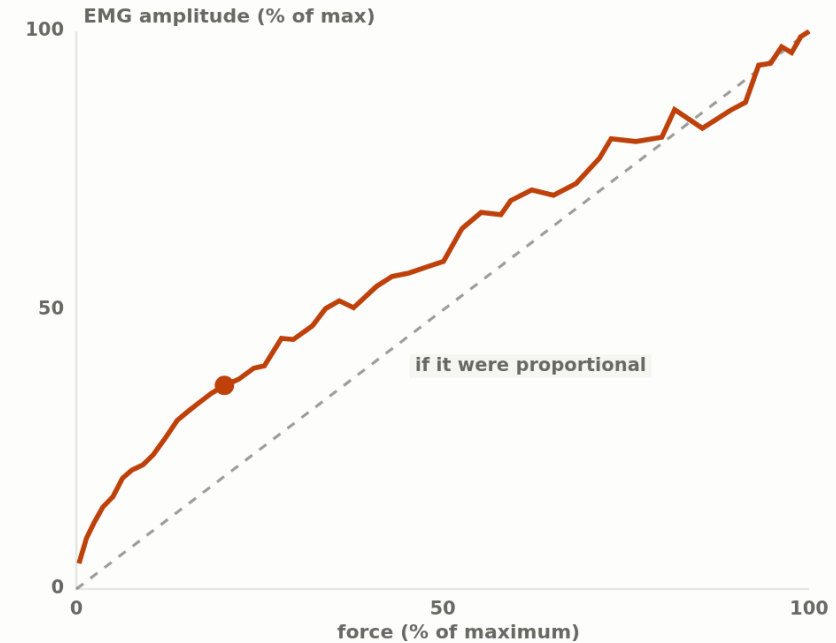


One electrode, many motor units

90 of 120 units active



500 ms of signal at one electrode



At **30 %** drive, **90** of 120 units are active: **20 %** of maximum force and **36 %** of maximum EMG — above the line of proportionality. Independent trains add in **power**, not amplitude, and the bend is set by the two assumptions on the right: how much louder a big unit is than a small one, and how far up the range recruitment continues. The shape of the EMG–force relationship is not a law.

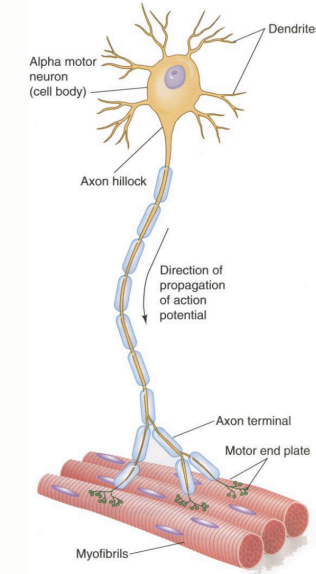
Neural drive **30 %**
Recruited by **85 %**

Size exponent **k = 0.50**

The electrochemical mechanism of contraction

- Neural signals travel along the motor axon and disperse through the fibres it innervates
- The signal propagates by the opening of **voltage-gated sodium channels** in the neurolemma and the sarcolemma

That propagating change in membrane potential is the thing EMG measures. Everything downstream of it — calcium, crossbridges, force — is invisible to the electrode.

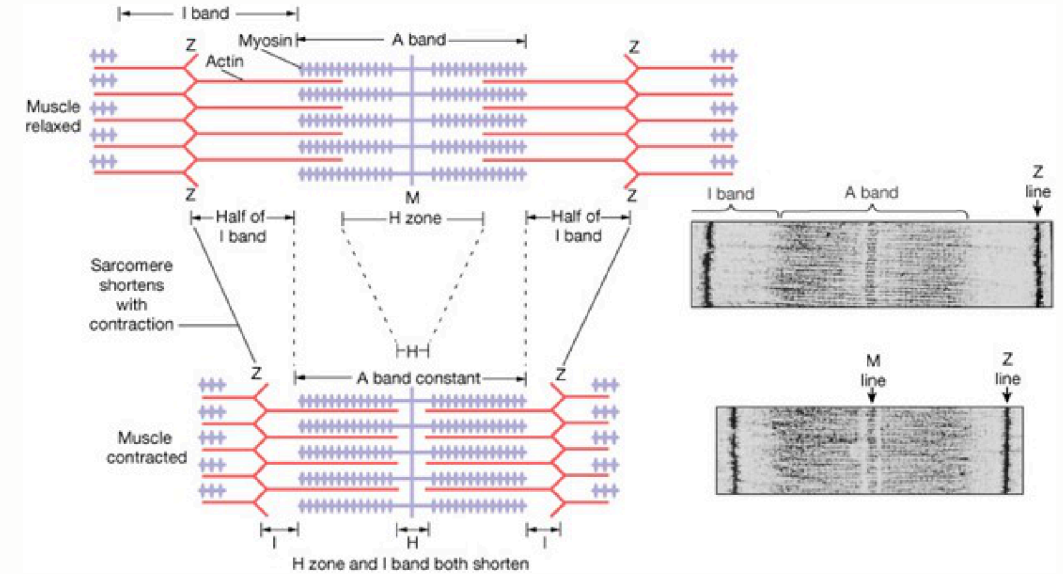


The action potential

Excitation–contraction coupling

Movement of the structural components of muscle is produced through the propagation of the action potential within the muscle, by a process called **excitation–contraction coupling**.

This is where the delay comes from. The electrical event finishes in a few milliseconds; the force takes tens of milliseconds to appear.



From the end plate to calcium release

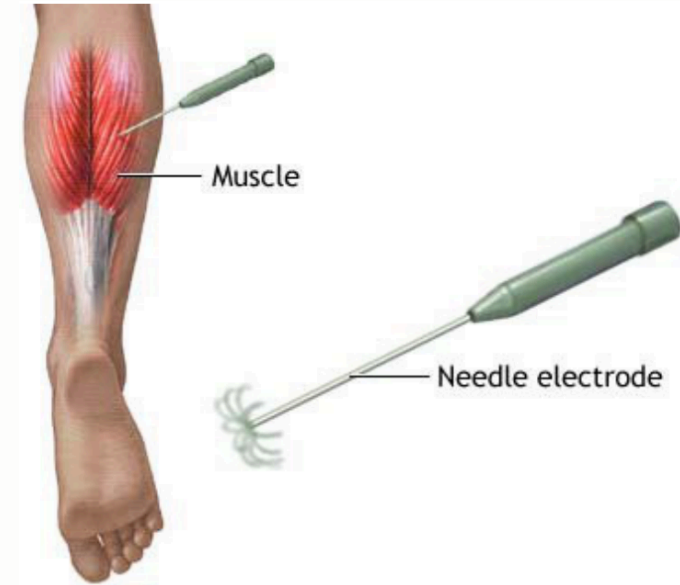
...and from calcium to the power stroke

Contraction, and a signal we can measure

We know the contraction of muscle is tied to an electrical event, so we can try to find a way to measure that electrical event.

Electromyography means placing electrodes **on** or **in** the muscle and measuring the action potentials travelling along the muscle fibres.

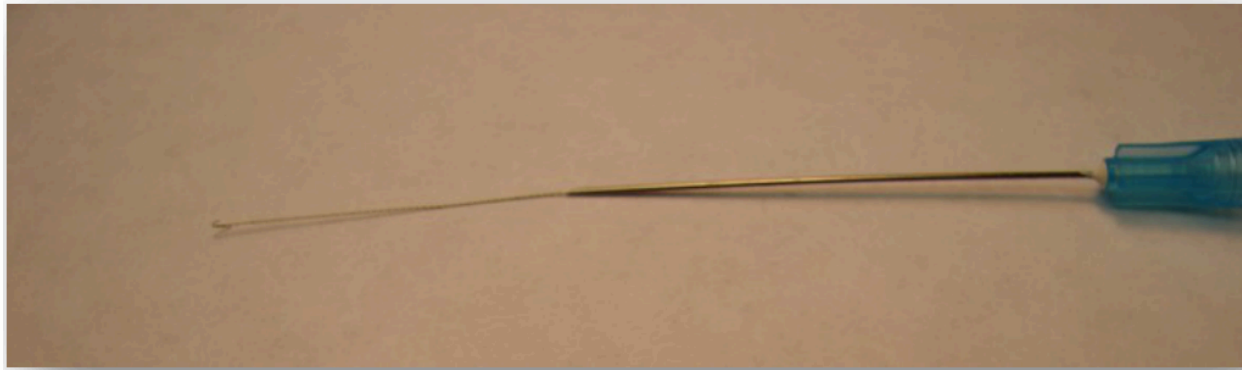
muscle → electrodes → amplifier → A/D converter
→ computer



Two ways of recording EMG

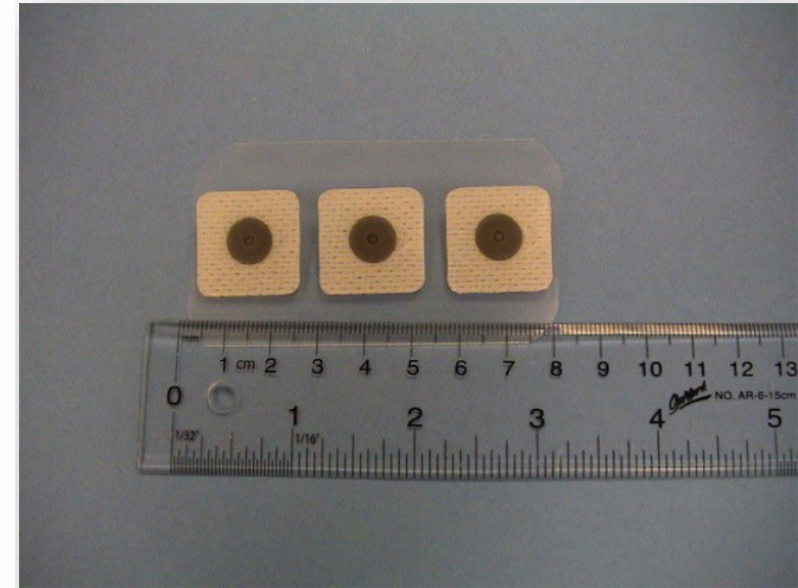
1. In-dwelling electrodes

fine wires or needles placed *in* the muscle



2. Surface electrodes

pairs of contacts placed *on* the skin over the muscle



In-dwelling electrodes

Advantages

record activity from a **single motor unit**

access to **deep** musculature

little **cross-talk**

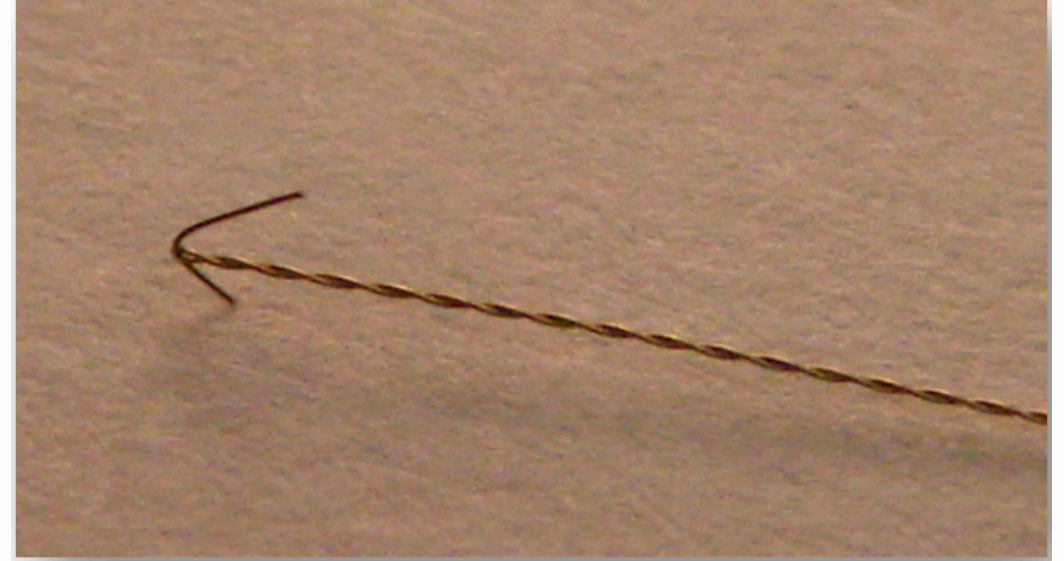
cross-talk: the unintentional recording of EMG from another muscle

Disadvantages

can be **painful** for participants

a single motor unit may not represent the whole muscle

limitations recording **dynamic** muscle activity



Surface electrodes

Advantages

easy to record

minimal discomfort to participants

indicates **global** muscle activity

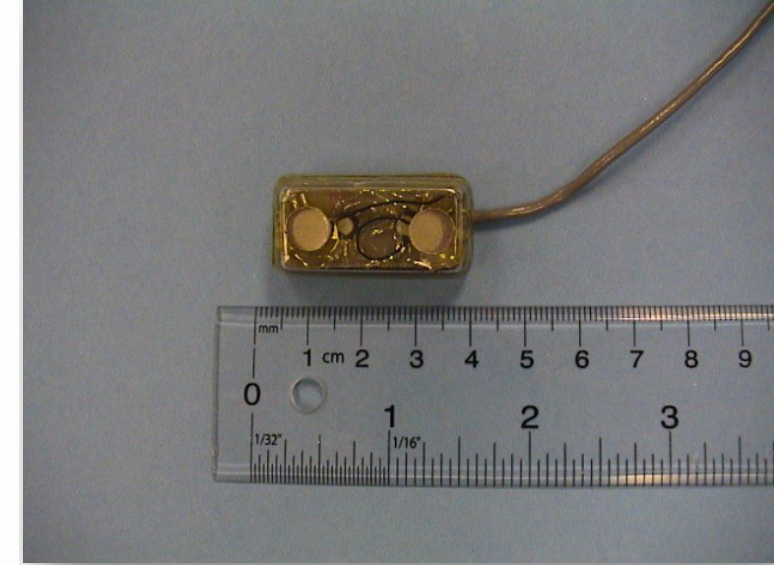
Disadvantages

only for **superficial** muscles

signal contaminated by **cross-talk**

may affect the movement patterns of the subject

Surface EMG is the algebraic **sum** of all motor unit firings near the electrode at that instant. It is a crowd, not a voice.

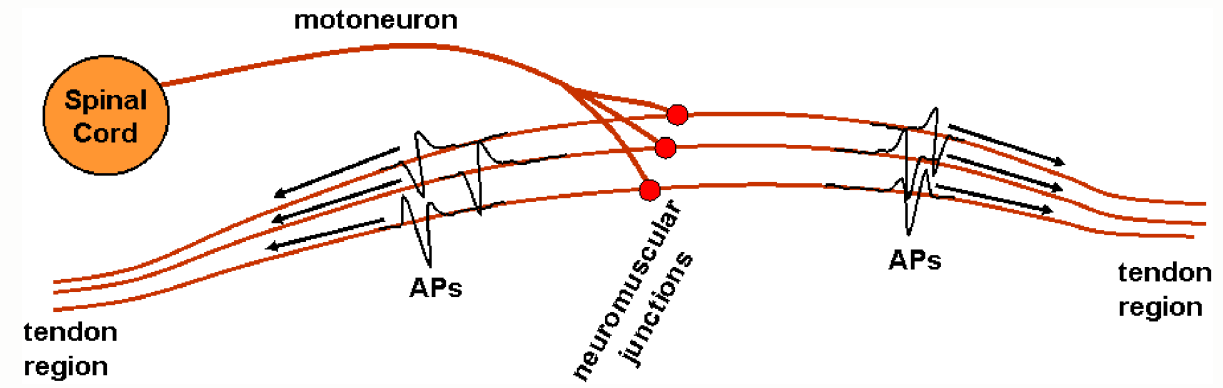


Electromyography

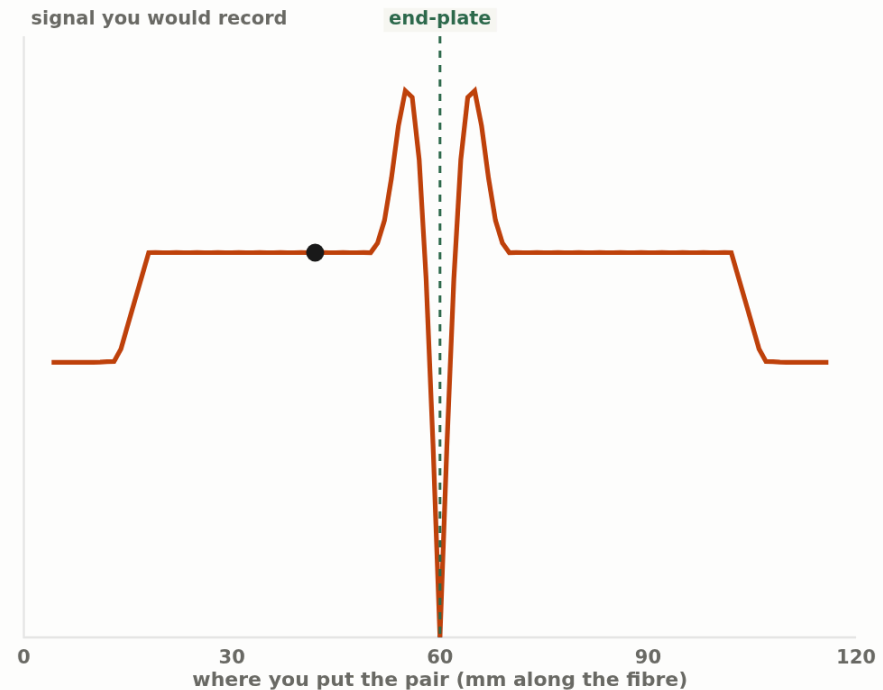
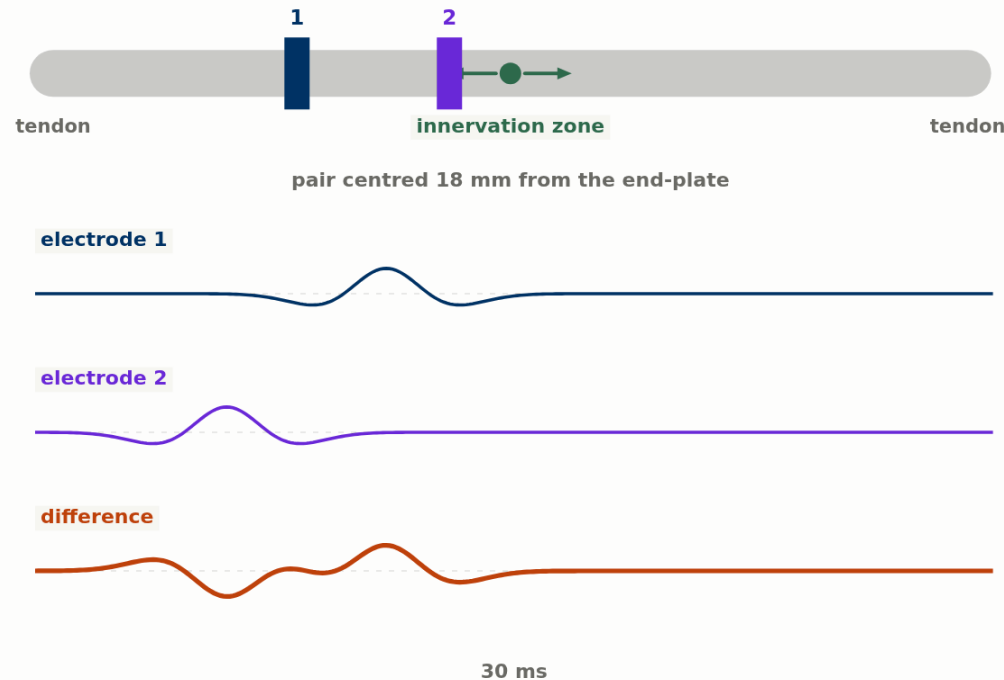
The action potential travels past the electrodes

- As action potentials travel along the muscle fibres, EMG electrodes — indwelling or surface — measure the **change in voltage** that accompanies their travel
- The potential is generated at the **motor end plate** and propagates towards **both** tendons

Because it travels, *where* you put the electrodes decides what you record. That is the next figure.



Where you put the pair, and what it costs you



The pair sits **18 mm** short of the end-plate and records **70 %** of the best signal available on this fibre. Astride the innervation zone the two electrodes see mirror images and the difference **cancels**; close to the tendon the travelling wave has run out of fibre. The two spikes flanking the null are a trap, not a target: there the amplitude swings from nothing to everything over a few millimetres, so a sensor that shifts slightly on the skin changes your answer. The flat plateau over the belly is the only stable place. Modelled: there is no electrode array in this dataset, so no conduction velocity here is measured.

Pair position

Electrode spacing

Innervation zone at

Conduction velocity

One electrode is very sensitive to things that are not muscle

A problem with **unipolar** recording is that it will pick up anything that changes the potential at that point, not just muscle:

- movement of the electrode or the cable
- electronics — computers, lighting, anything on the mains
- other muscles nearby

These are referred to collectively as **noise**.



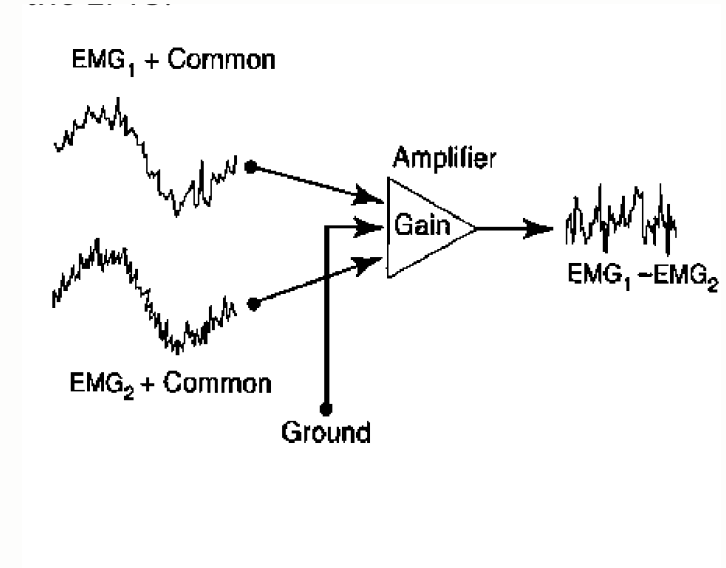
Two electrodes, and a differential amplifier

- Use a **bipolar** (two-pole) configuration and a **differential amplifier**
- The signal and the noise are sampled at **two** points; subtract one from the other and what is *common* to both cancels

$$(m_1 + n) - (m_2 + n) = m_1 - m_2$$

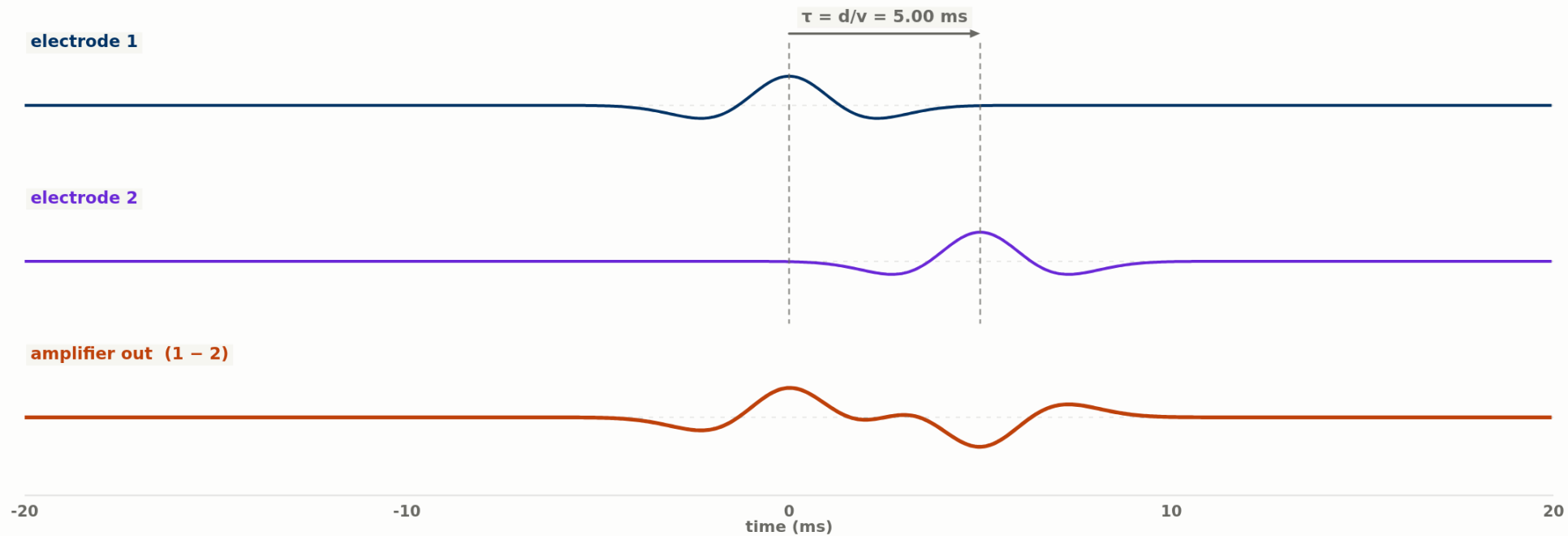
The result, $m_1 - m_2$, is the **motor unit action potential** (MUAP) as recorded: the electrical activity of all the fibres in one motor unit, seen as a difference.

after Enoka 2002, Fig. 1.30

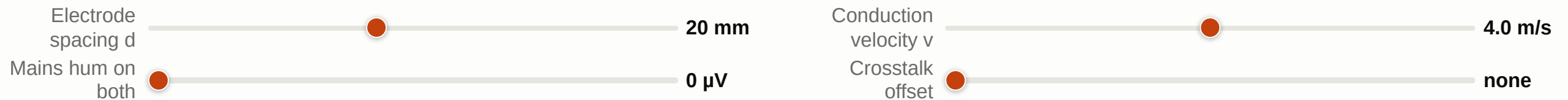


Enoka 2002 Fig 1.30

What the difference actually looks like



The two electrodes see the same potential **5.00 ms** apart, so the difference is biphasic. Differencing is a comb filter with its first zero at $v/d = 200 \text{ Hz}$ — **inside** the 20–450 Hz EMG band.



The two signals

Add the spectrum

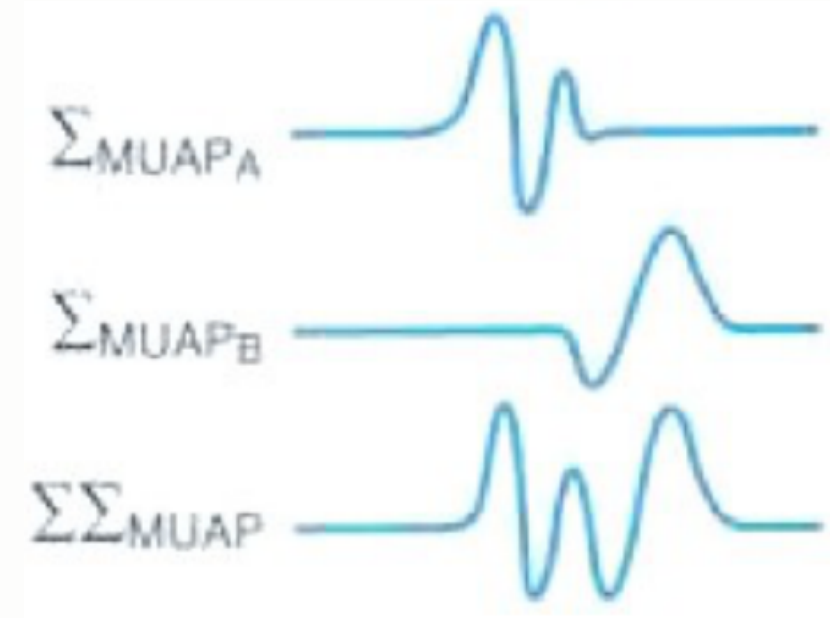
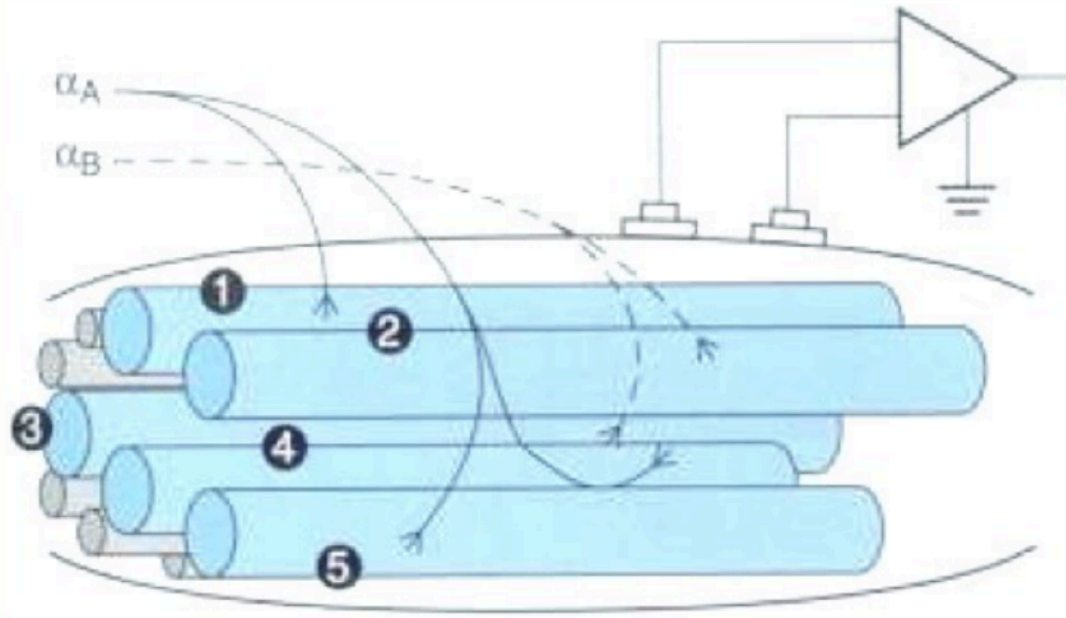
What differencing removes — and what it does not

Removed. Anything genuinely *common* to both electrodes: mains hum, and any distant source far enough away that it arrives at both contacts identically. The cancellation is exact.

Not removed. Cross-talk. A neighbouring muscle is *nearer one electrode than the other*, so it does not arrive identically and does not cancel. In the figure opposite, a source sitting just a few millimetres off centre survives the subtraction at about a **quarter** of the amplitude of the signal you wanted.

So a clean-looking trace is not evidence that you are recording the muscle you think you are. Differential amplification solves the noise problem. It does not solve the **anatomy** problem — that is what electrode placement is for.

Every motor unit has its own signature, and they all add up

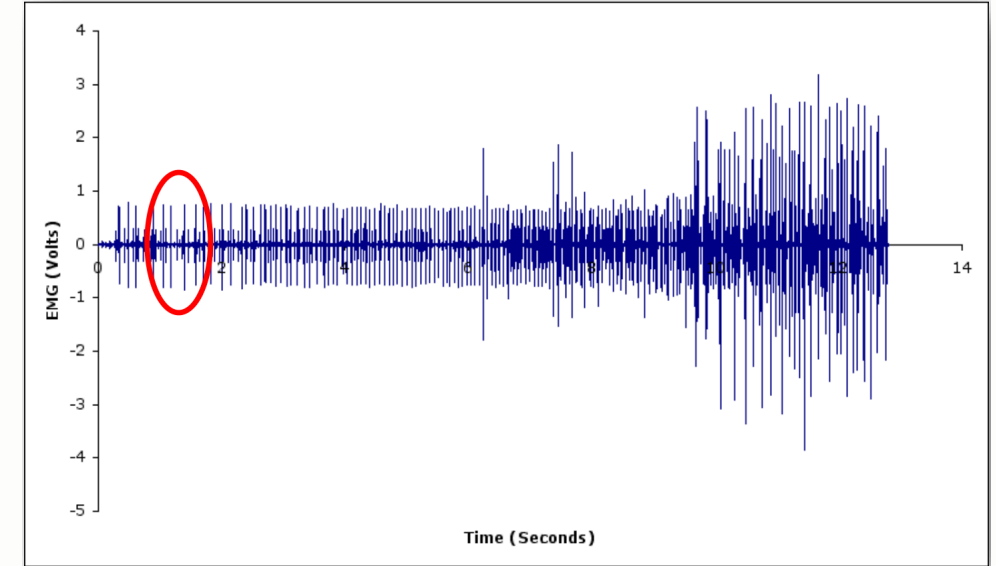


- Each motor unit has its own characteristic MUAP, set by the number, type and position of its fibres relative to the electrode
- Very rarely is one motor unit firing alone — the electrode records the **sum** of many units firing at different times

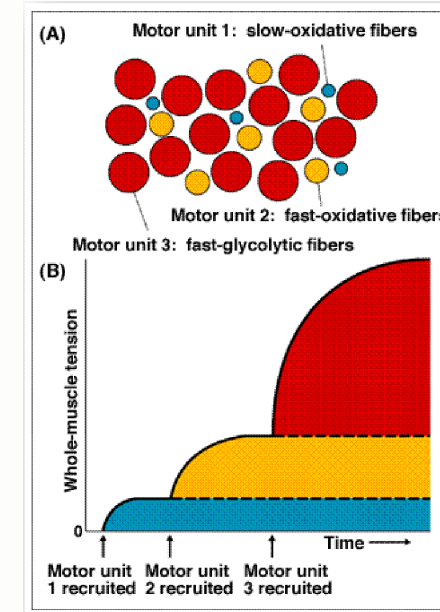
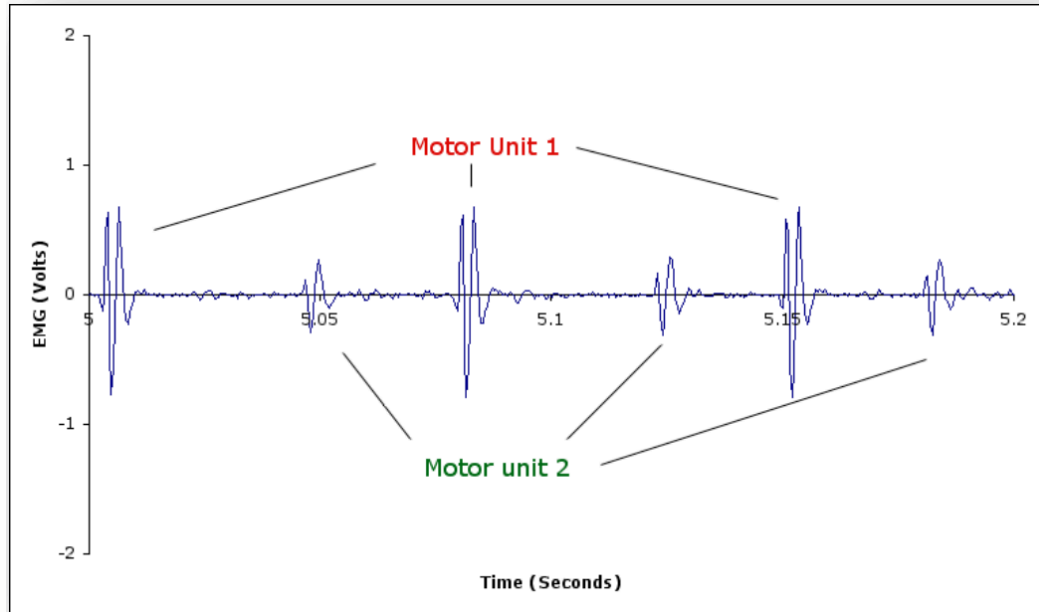
The interference pattern

The surface EMG signal is therefore a **summation of all the MUAPs** near the electrode, which gives a gross approximation of the activity of the whole muscle — useful for **timing, rate of force development** and **fatigue**.

Once the units overlap you cannot count them any more. That is why surface EMG answers “when” and “roughly how much” rather than “which units”.



Henneman, seen on the electrode

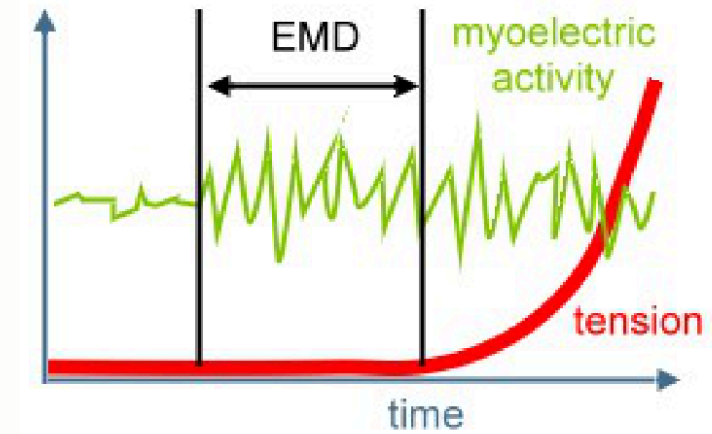


During a contraction the small motor units are normally recruited first and the larger ones after — orderly recruitment from small to large is the **Henneman size principle**. It gives smooth force production and preferentially uses the non-fatigable units, so the muscle resists fatigue.

Some other important considerations

- **Electrode placement**
 - the neuromuscular junction — the motor end plate
- **EMG signal processing**
- **Electromechanical delay**

The rest of the lecture is these three, in that order.

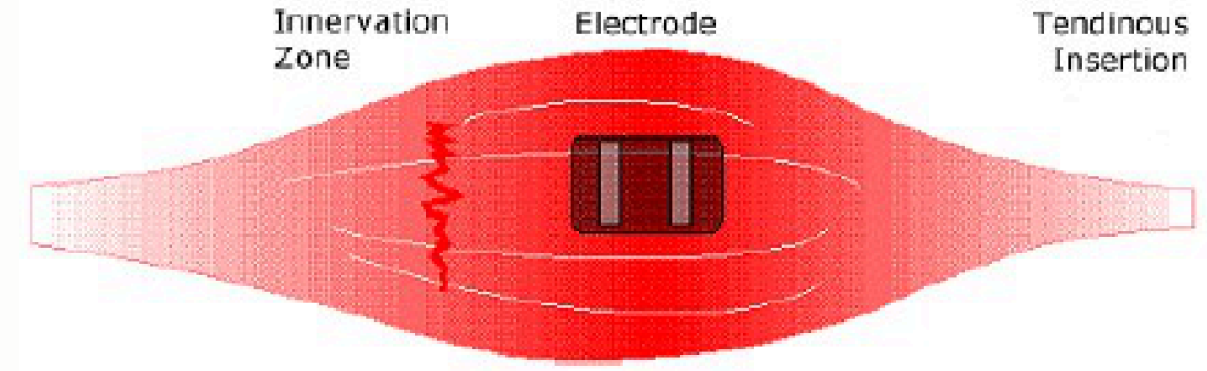


Electromechanical Delay (EMD)

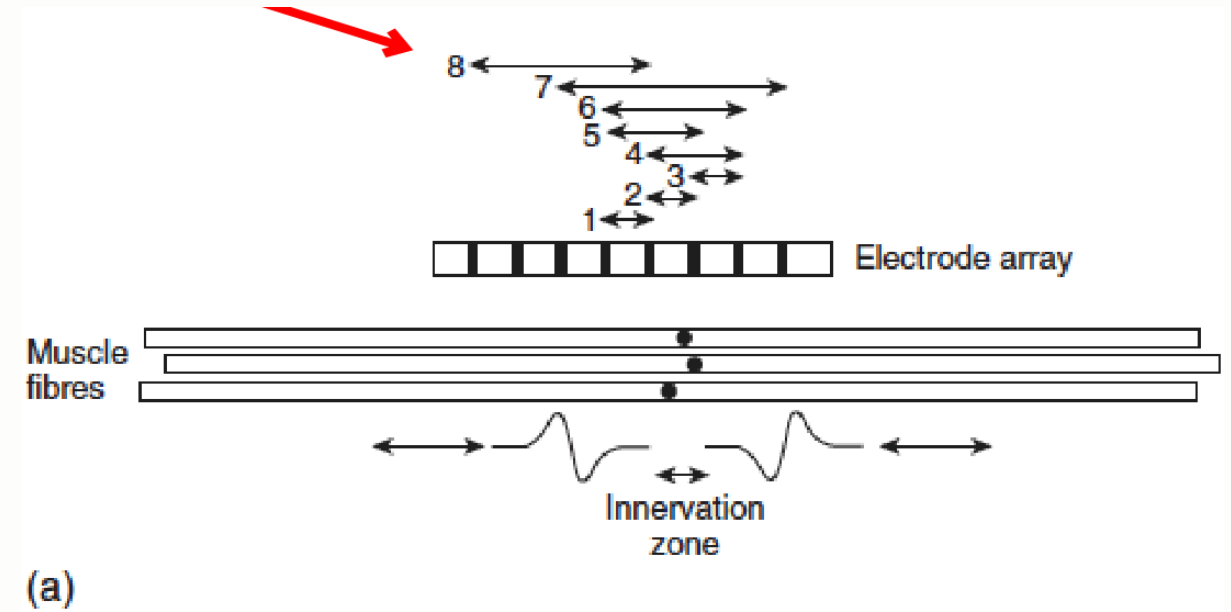
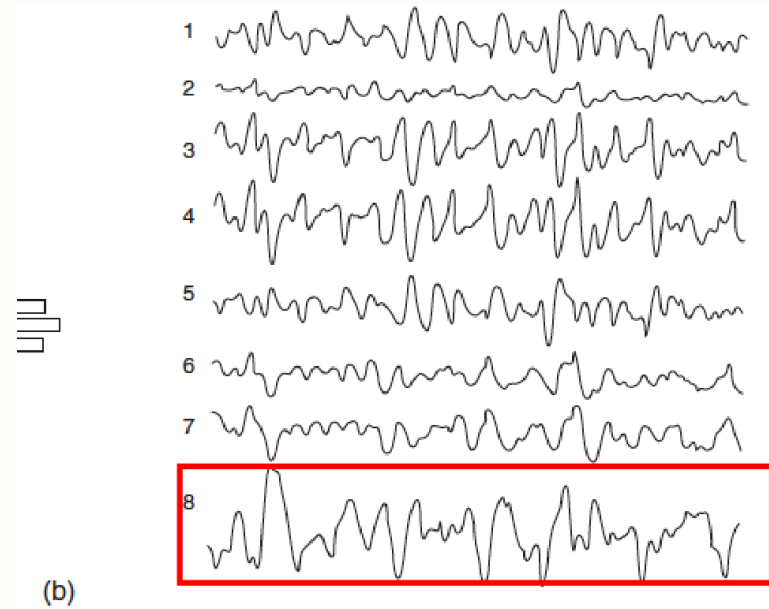
Orienting the electrodes on the muscle

- Orient the electrodes so the MUAPs travel **parallel** to the line joining them — along the fibres, not across them
- Place them on the **muscle belly**, and **not** on the innervation zone (the motor end plates)

Across the fibres there is no delay between the two contacts, so the difference has nothing to work with and the signal collapses — the same arithmetic as sitting on the innervation zone.



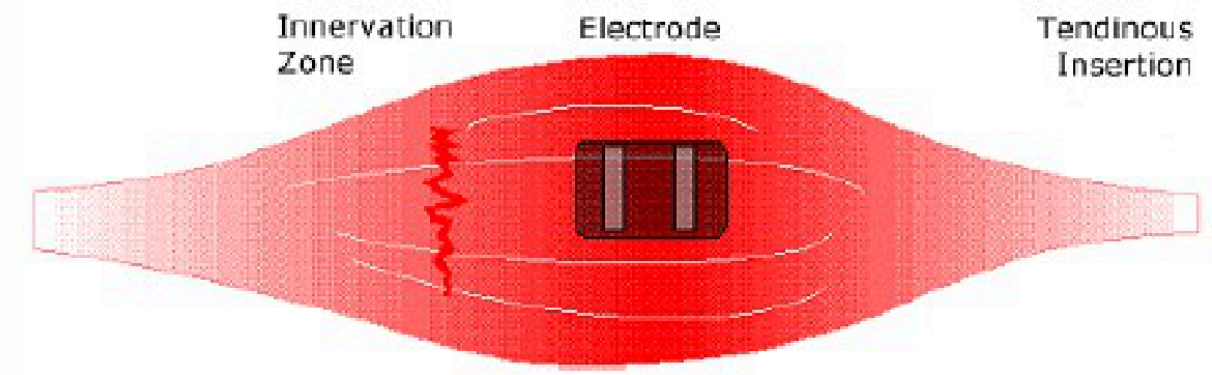
The evidence, measured along a real muscle



Payton and Bartlett, *Biomechanical Evaluation of Movement in Sport and Exercise*, 2008

Electrode placement summary

- On the **muscle belly**
- Avoid the **tendon** and the lateral edge of the muscle
- Align the electrodes **parallel with the muscle fibres**
- Mind the **motor point** or innervation zone
 - place on the midline of the belly, midway between the tendon and the nearest innervation zone
 - avoid placing directly over the motor point

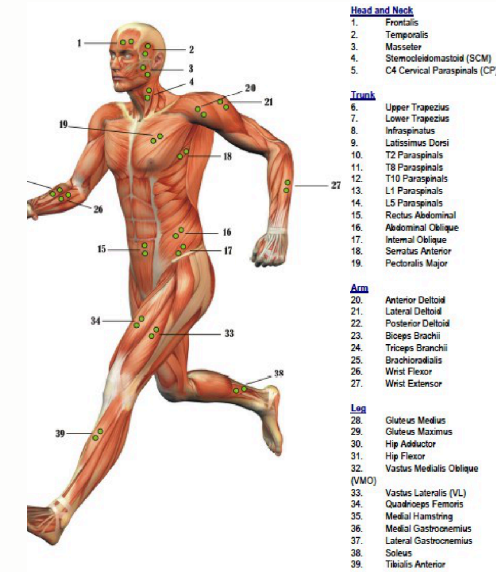


Standard placements — SENIAM

You do not have to work the placement out from first principles every time. **SENIAM** (Surface EMG for the Non-Invasive Assessment of Muscles) publishes an agreed position and orientation for each muscle.

www.seniam.org

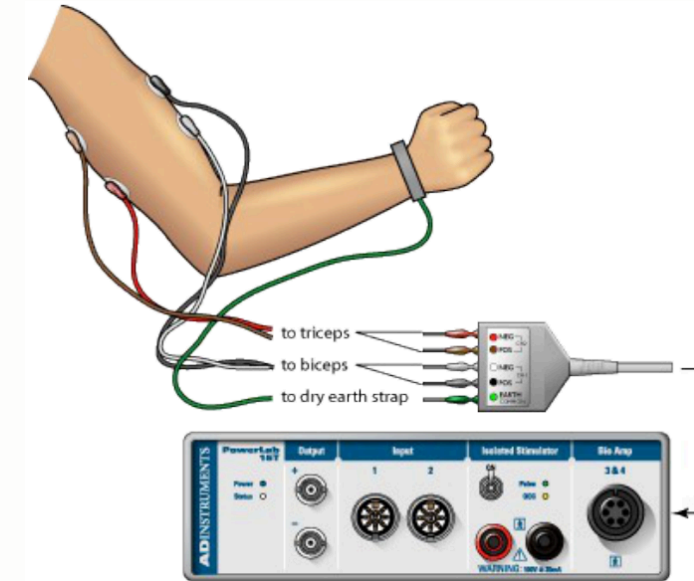
Using a standard placement is what makes your numbers comparable with anyone else's — including your own, next week.



Setting up — preparing the skin

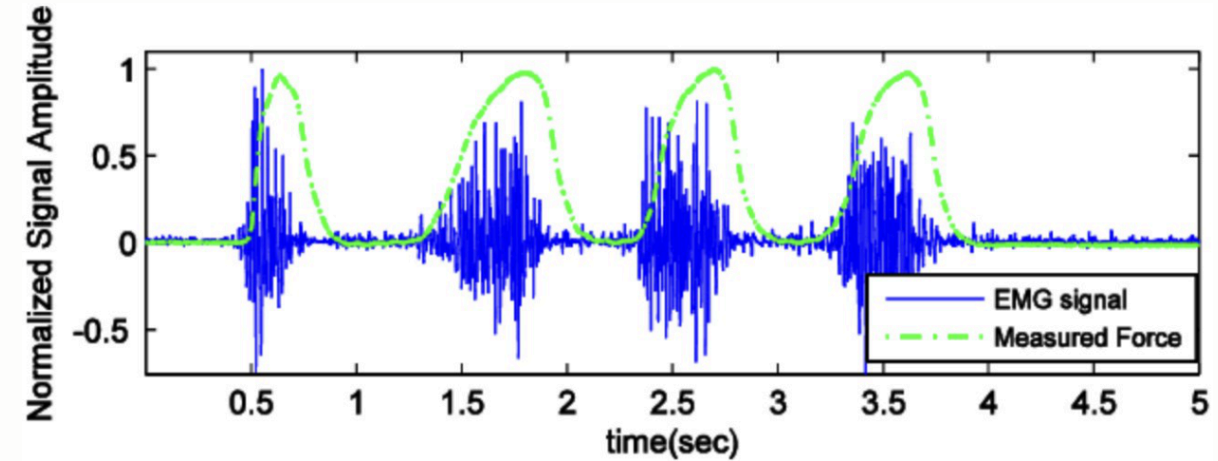
- Make sure the site is properly **cleaned and prepared**
- Tissue — skin and subcutaneous fat — acts as a **filter**
- Clean the skin with **alcohol**
- Shave excess hair
- Lightly abrade the skin
- Use electrode gel, paste or cream

Skin impedance is the one part of the signal chain you can improve for free, before you record anything.



Why process the signal at all

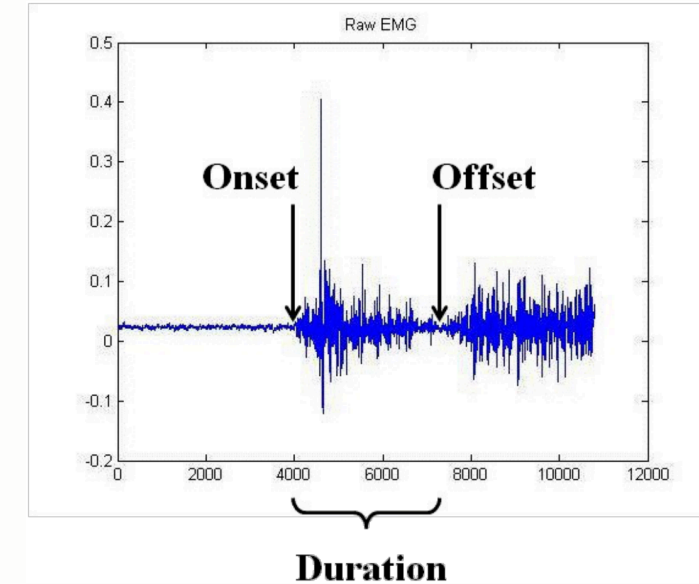
- EMG is an **indirect** measure of muscle force
- It also carries neural information — **which** motor units fire and **when**
- How we process it decides what we can read from it:
 - **timing** of muscle activity
 - **force**
 - **fatigue**



Timing: onset and offset

- The raw EMG can be used **without any processing** to tell us about timing — when the muscle turns on and off
- Processing usually improves the result, but it is not required for timing

Normalisation. Amplitude must be made *relative* before it can be compared between people or conditions:
 $\text{normalised EMG} = \text{EMG} \div (\text{max EMG})$, usually from an **MVC** or MVIC.

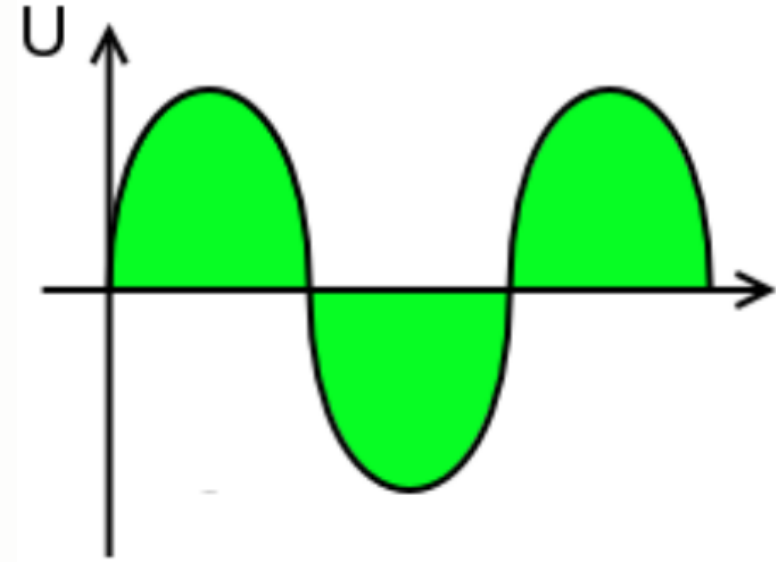


Full-wave rectification

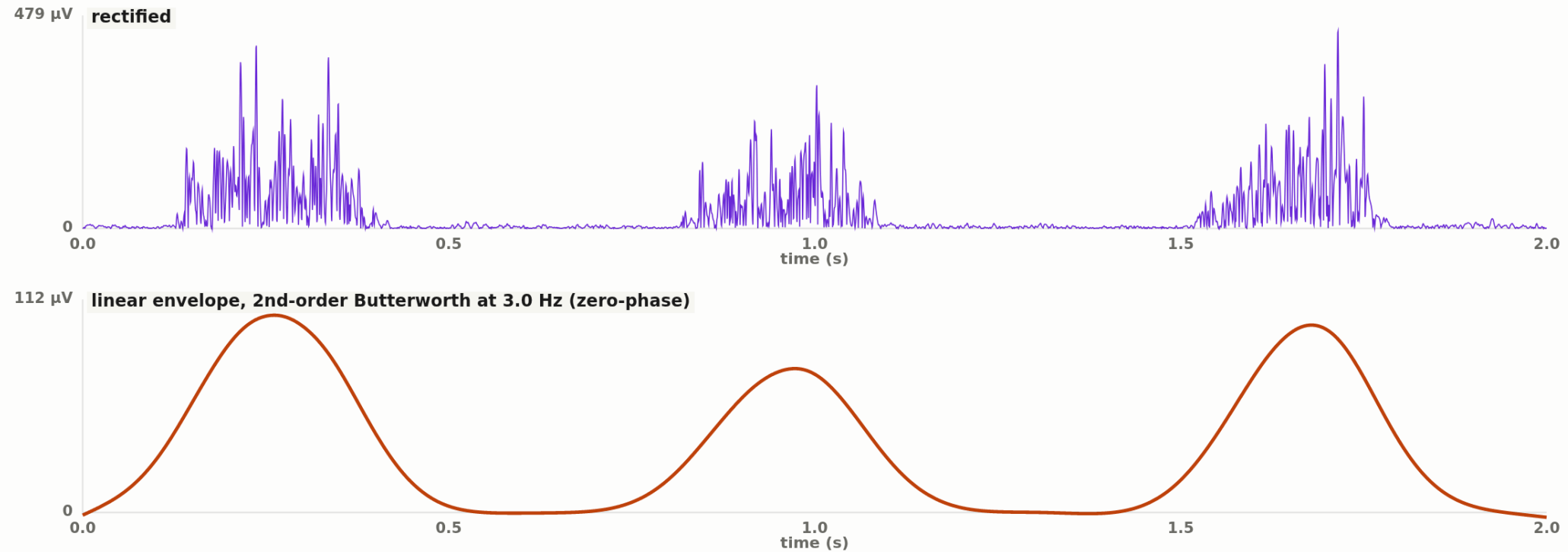
The most common way to process EMG is to produce an average value associated with its magnitude. Two steps:

1. Take the **absolute value** of the signal — full-wave rectification (FWR)
2. Take an **average** over some time period

Step one is not optional. Raw EMG is **zero-mean** by construction, so averaging it without rectifying gives you nothing but zero.



The processing chain, on a real recording



A low-pass filter on the rectified signal gives the **linear envelope**. Filtering forwards and backwards (**filtfilt**) leaves the envelope in time with the burst that made it.

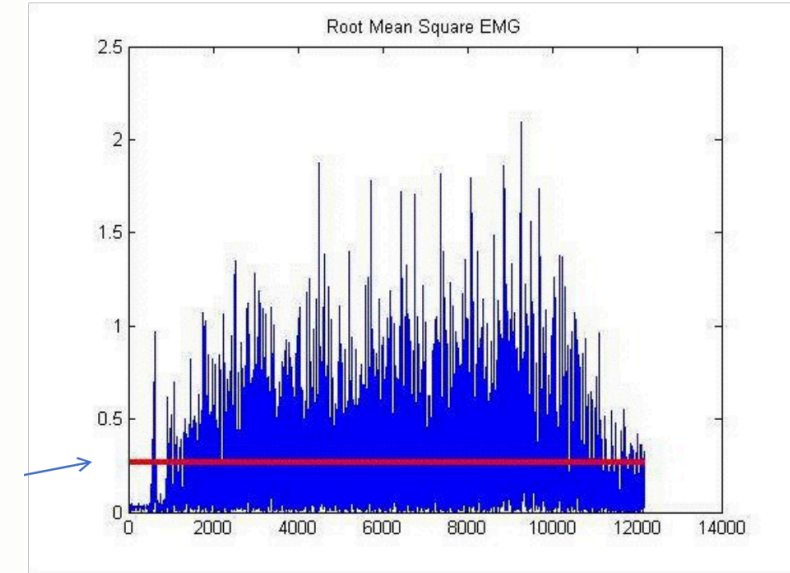


Root mean square

Taking the **root mean square** is a way of taking an average magnitude of the EMG signal. It can be used over different time windows.

$$\text{RMS} = \sqrt{\left(\frac{1}{T} \cdot \int_t^T \text{EMG}(t)^2 dt \right)}$$

Squaring removes the sign, so RMS does not need a separate rectification step — and unlike the rectified mean it has a physical meaning: it is the signal's **power**.

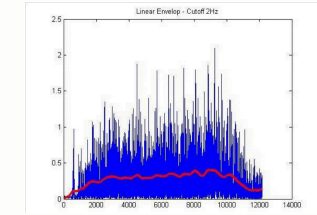
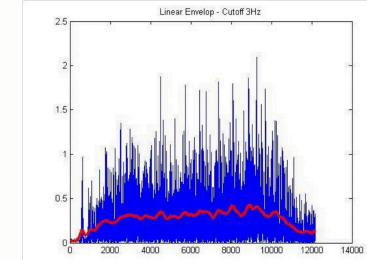


The linear envelope

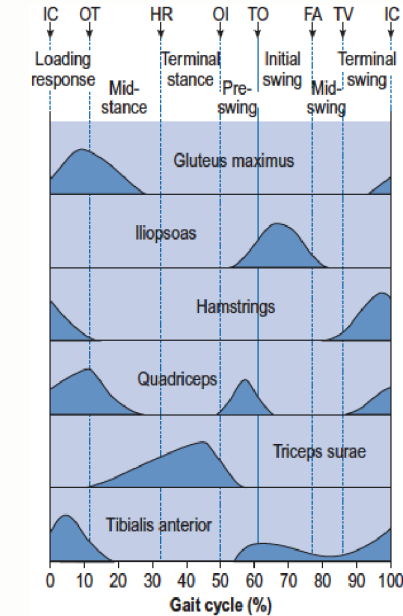
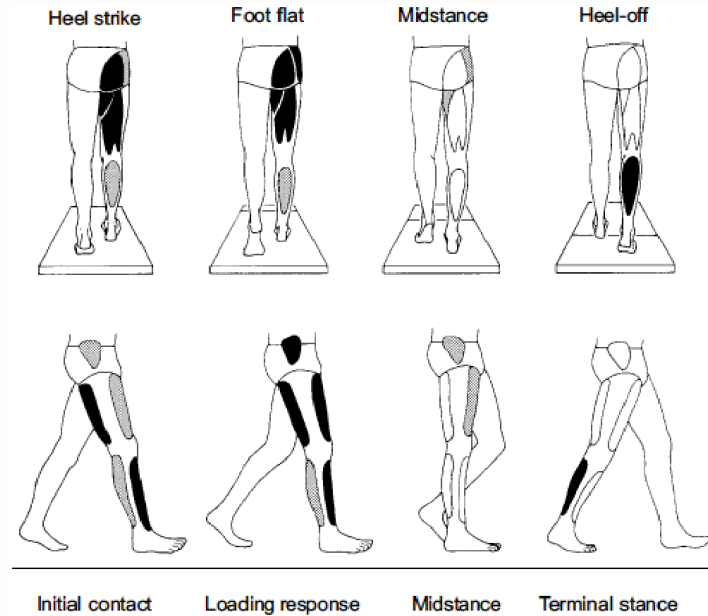
Another way to average is to **low-pass filter** the rectified signal. The resulting tracing is called a **linear envelope**.

The filter can be set to be more or less sensitive. A lower cut-off is smoother and slower; a higher cut-off follows the bursts more closely and is noisier.

Filter **forwards and backwards** (zero-phase) or the envelope comes out late — and you will read that lag as physiology.



Putting it to work: EMG through the gait cycle

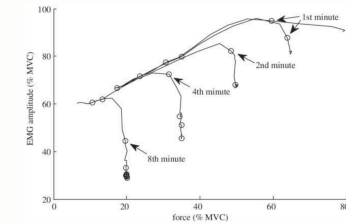
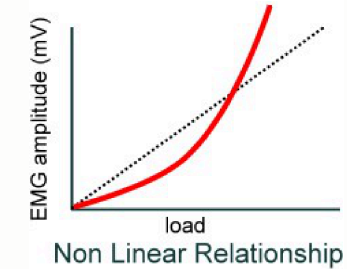


Processing the EMG this way lets us quantify and visualise muscle activity during a task, with respect to both **timing** and **amplitude**.

EMG and force — the relationship is not a straight line

- There is a **non-linear** relationship between EMG and muscle force, even during **isometric** contractions
- During **fatigue** the relationship changes
- During **dynamic** contractions it becomes more complex still — length and velocity are changing, and so is the tissue between muscle and electrode

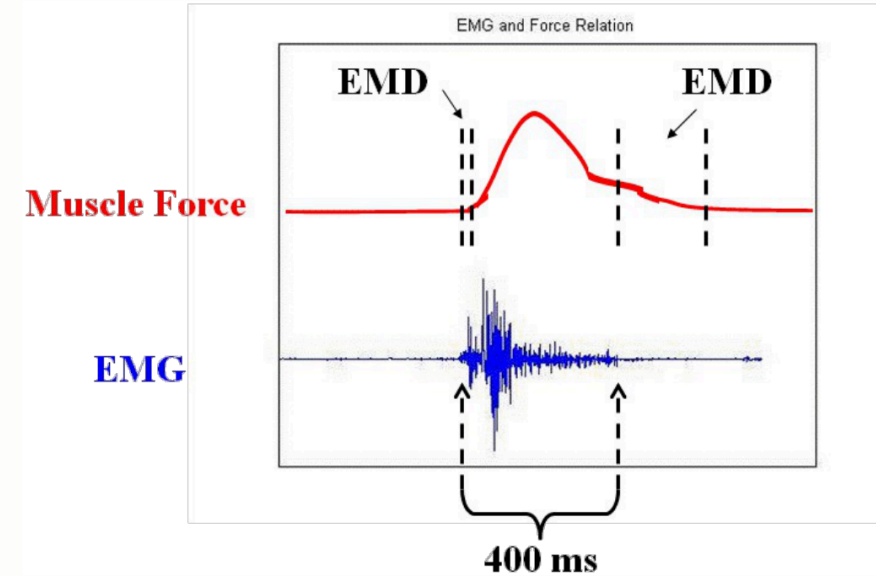
We derived the reason earlier: independent trains add in **power**, and the units are not all the same size. The shape of the curve follows from recruitment, not from a law of nature.



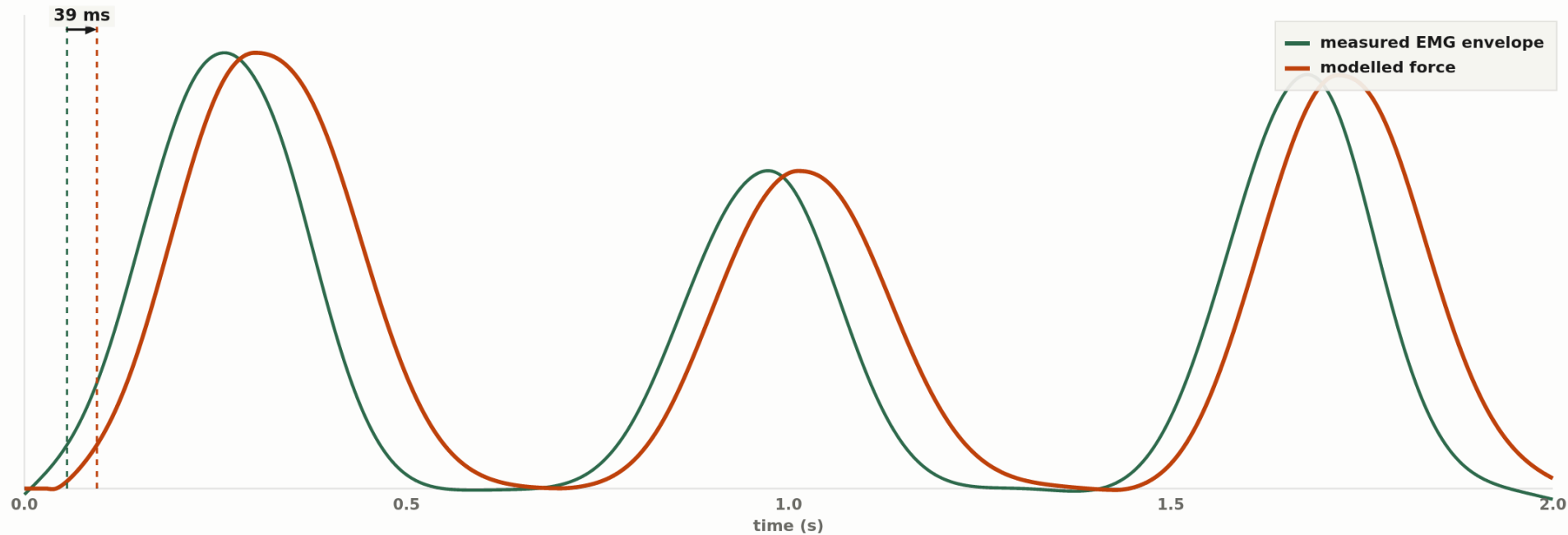
Electromechanical delay

- The delay between **neural activation** of a muscle and the development of **muscle tension**
- Approximately **30–50 ms**, depending on the muscle and the joint
- It comes from excitation–contraction coupling and from taking up slack in the **series elastic** elements

Relating EMG to force or to movement without allowing for this will misplace both.



Why force arrives after the signal



Force onset trails EMG onset by **39 ms** here: **30 ms** of electromechanical delay plus the time the first-order activation takes to climb past the threshold. Note what the two time constants do at the END of each burst — deactivation is the slower of the two, so force outlasts the signal that caused it. Any measurement that lines EMG up with force without allowing for this will misplace both. Envelope measured; force modelled.

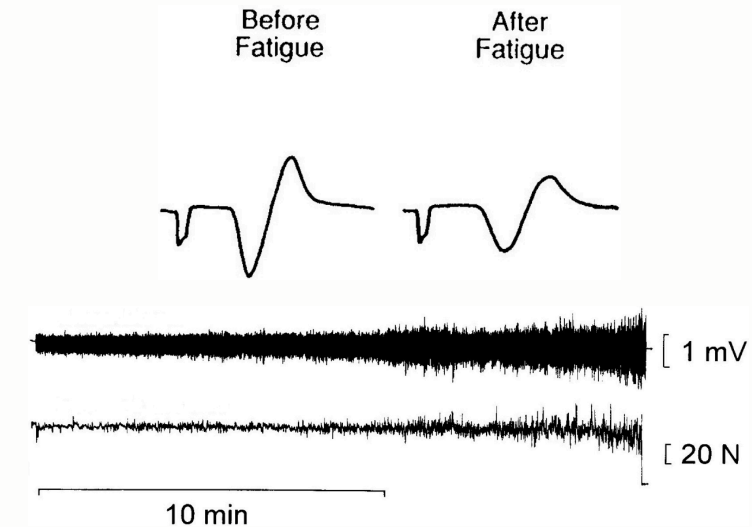
Electromechanical delay
Deactivation τ

Activation τ

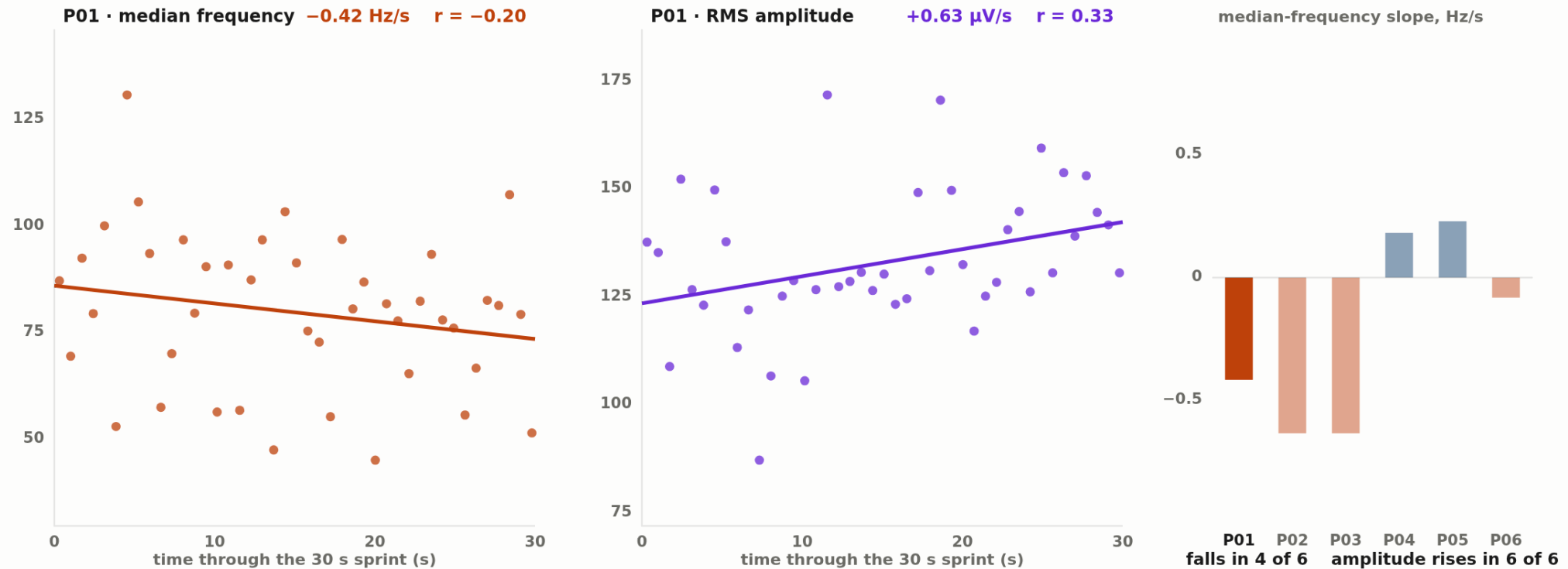
Fatigue — what the textbook says

- A single MUAP **decreases** in amplitude with fatigue, and produces less force
- Surface EMG of a submaximal contraction **increases** in amplitude as fatigue develops, because more motor units are recruited to hold the same force
- The frequency content **shifts down**: conduction velocity falls, so each MUAP is stretched out in time and the spectrum is compressed

Two predictions: amplitude up, median frequency down. The next slide tests both against measured data.



...and what six real sprints show



Vastus lateralis through a 30 s all-out sprint, one point per pedal revolution. For **P01** the median frequency goes **down** 13 Hz over the sprint ($r = -0.20$) while amplitude goes **up** ($r = 0.33$). Across all 6 participants amplitude rises in **6** of them but median frequency falls in only **4**. The frequency shift is a tendency, not a law — it is far more reliable in a sustained isometric hold than in dynamic cycling, where the muscle is only active for part of each revolution and the detection volume moves under the skin.

P01

P02

P03

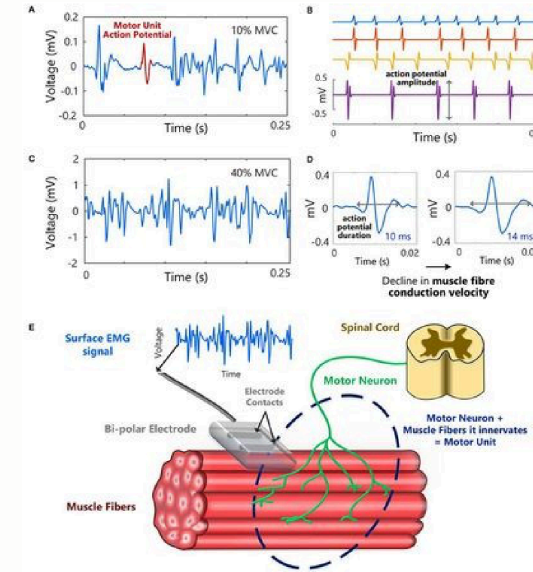
P04

P05

P06

What the signal can be analysed for

- **Coordination**
 - sequencing and co-contraction
- **Force**
 - linear and non-linear relations — with care
- **Fatigue**
 - decrease in the single MUAP
 - increase in surface EMG during a fatiguing contraction
 - change in the frequency content



What to take away

- **EMG is the electrical event, not the force.** Everything between the action potential and the crossbridge is invisible to the electrode.
- **Bipolar recording removes what is common.** Mains hum goes exactly; cross-talk is not common to both electrodes and stays.
- **Placement is arithmetic, not folklore.** On the end plate the difference cancels; at the tendon the wave runs out; the belly is the stable plateau.
- **Normalise before comparing.** Raw millivolts are not comparable between people, days, or electrode placements.
- **It is a sum, and an algebraic one.** Potentials cancel as well as add, so amplitude under-reports at high force.
- **Spacing is a filter.** Differencing two contacts d apart has zeros at multiples of v/d — 20 mm at 4 m/s notches 200 Hz, inside the band.
- **Process deliberately.** Rectify because the raw mean is zero; filter forwards and backwards or you will read your own lag as electromechanical delay.
- **Amplitude rises with fatigue; frequency usually falls.** “Usually” is the honest word — 4 of our 6 sprinters, against 6 of 6 for amplitude.

BIOMECHANICS (EPHE 341)

Electromyography

Next up: angular kinetics · Measured data:
Cartier et al., six participants, leg cycling